

Assignment : Op-Amp Modeling, Feedback Networks and Real Op-Amp Configurations

Problem 1 : OP-amp Modeling and feedback network

1. What kind of configuration shown in the following figure?
2. Calculate the V_o/V_i for the following cases
 - a. When $r_d = \infty$, $R = 0$, $V_i = 1V$
 - b. When $r_d = 2M\ \Omega$, $r_o = 75\ \Omega$, $V_i = 1V$
 - c. Compare the both cases : a vs b and write the conclusion
 - d. Model the EOA using VCVS in LT Spice and plot the V_o/V_i by varying the input signal from 1V, to 5 V and then tweak the value of r_d to observe the changes in V_o/V_i

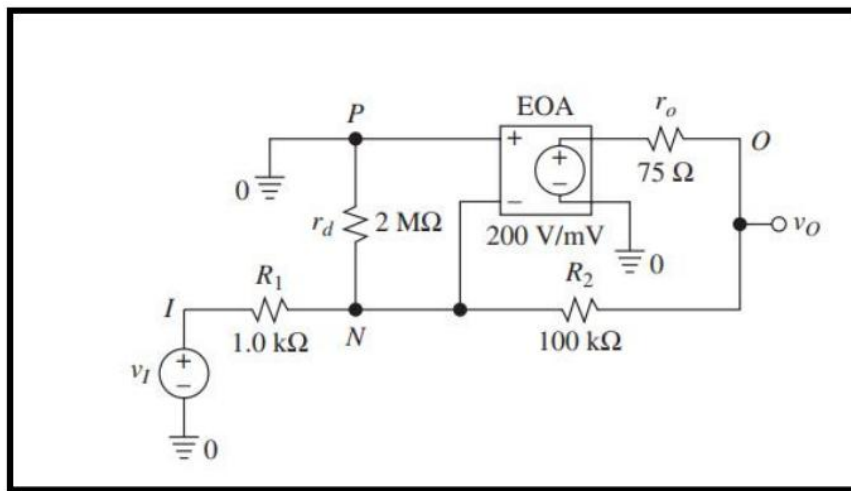


Fig 1: Problem 1 - Inverting amplifier circuit with EOA model ($r_d = 2M\Omega$, $R_1 = 1k\Omega$, $R_2 = 100k\Omega$, $r_o = 75\Omega$, $A_{ol} = 200\ V/mV$)

Solution :-

The configuration shown in Fig. 1 is an Inverting Amplifier. The input signal V_i is applied through R_1 ($1k\Omega$) to the inverting (-) input of the op-amp, with feedback resistor R_2 ($100k\Omega$) connected from output back to the inverting input. The non-inverting (+) input is grounded through r_d .

Case A: $r_d = \infty$, $R_0 = 0$, $V_i = 1V$ (Ideal Op-Amp)

With $r_d = \infty$ (ideal input resistance) and $R_0 = 0$ (ideal output resistance), the op-amp behaves as an ideal device with negative feedback applied (V4C). Under ideal op-amp conditions (virtual short and virtual open):

$$V_- = V_+ = 0 \text{ (virtual ground at inverting input)}$$

$$V_o/V_i = -(R_2/R_1) = -(100k/1k) = -100$$

$$V_{out} = -100 \times V_i = -100 \times 1V = -100V$$

Hand Calculations :-

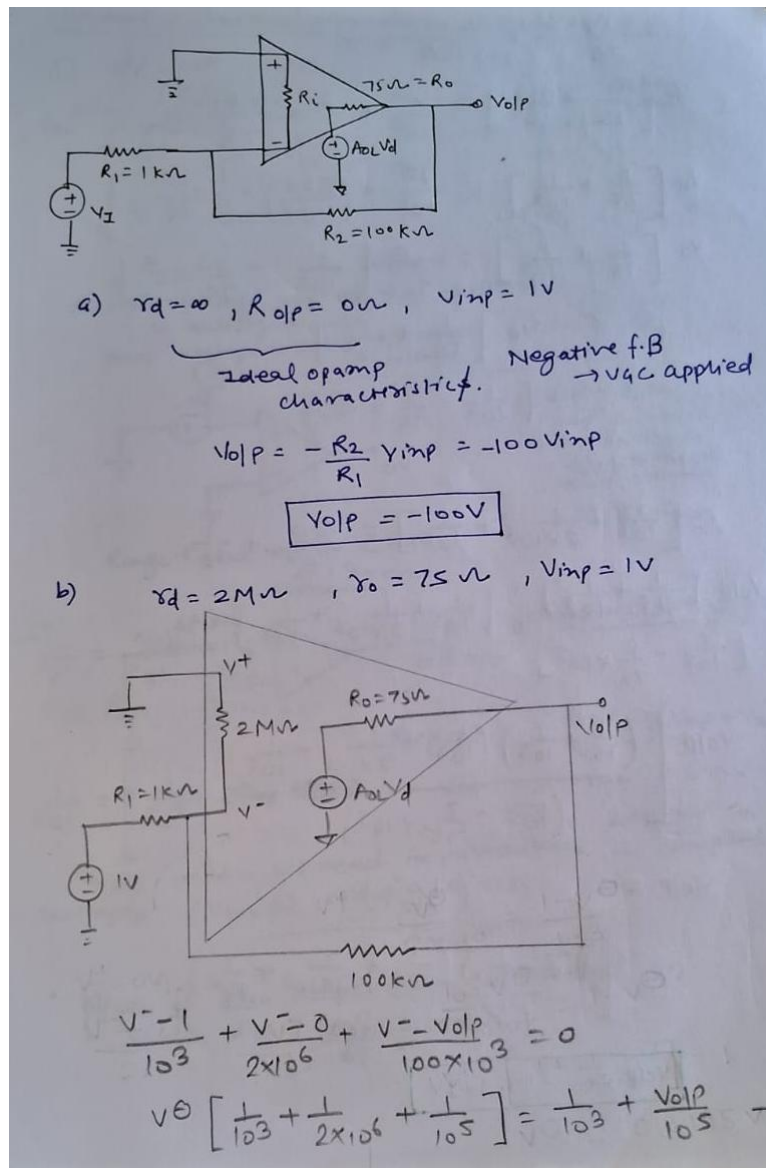


Fig 2: Hand calculation - Case A (ideal op-amp, $V_{out} = -100V$) and Case B circuit setup with KCL equations at node V^-

$$\frac{V_o - A_{OL}V_d}{75} + \frac{V_o - V^\ominus}{10^5} = 0$$

$$V_o \left[\frac{1}{75} + \frac{1}{10^5} \right] = \frac{V^\ominus}{10^5} + \frac{A_{OL} \times (0 - V^\ominus)}{75}$$

$$V_o \left[\frac{1}{75} + \frac{1}{10^5} \right] = \frac{V^\ominus}{10^5} + \left(\frac{200}{10^3} \right) \times \left(\frac{-V^\ominus}{75} \right)$$

$$V_o \left[\frac{1}{75} + \frac{1}{10^5} \right] = V^\ominus \left[\frac{1}{10^5} - \frac{2}{75} \times 10^5 \right]$$

$$V^\ominus = \frac{V_o \left[\frac{1}{75} + \frac{1}{10^5} \right]}{\left[\frac{1}{10^5} - \frac{2}{75} \times 10^5 \right]} \quad \text{--- (2)}$$

From eq (1)

$$V^\ominus \left[\frac{1}{10^3} + \frac{1}{2 \times 10^6} + \frac{1}{10^5} \right] = \frac{V_{oIP}}{10^5} + \frac{1}{10^3}$$

$$\frac{V_{oIP} \left[\frac{1}{75} + \frac{1}{10^5} \right]}{\left[\frac{1}{10^5} - \frac{2}{75} \times 10^5 \right]} \times \left[\frac{1}{10^3} + \frac{1}{2 \times 10^6} + \frac{1}{10^5} \right] - \frac{V_{oIP}}{10^5} = \frac{1}{10^3}$$

$$V_{oIP} \left[\frac{\left(\frac{1}{75} + \frac{1}{10^5} \right) \left(\frac{1}{10^3} + \frac{1}{2 \times 10^6} + \frac{1}{10^5} \right)}{\left(\frac{1}{10^5} - \frac{2}{75} \times 10^5 \right)} - \frac{1}{10^5} \right] = \frac{1}{10^3}$$

$$V_{oIP} = \frac{\left(\frac{1}{10^3} \right)}{\left[\frac{\left(\frac{1}{75} + \frac{1}{10^5} \right) \left(\frac{1}{10^3} + \frac{1}{2 \times 10^6} + \frac{1}{10^5} \right)}{\left(\frac{1}{10^5} - \frac{2}{75} \times 10^5 \right)} - \frac{1}{10^5} \right]}$$

$$V_{oIP} = -99.949V$$

Fig 3: Hand calculation – Case B full derivation yielding $V_{out} = -99.949V$

Case B: $r_d = 2M\Omega$, $r_o = 75\Omega$, $V_i = 1V$ (Practical EOA Model)

With finite input resistance $r_d = 2M\Omega$ and output resistance $r_o = 75\Omega$, KCL is applied at nodes V^- and V_o simultaneously. Writing KCL at the inverting node (V^-) and at the output node, then solving the two simultaneous equations:

From equation (1) - KCL at V^- :

$$V - [(1/10^3) + (1/2 \times 10^6) + (1/10^5)] = V_{out}/10^5 + 1/10^3$$

From equation (2) - KCL at V_{out} :

$$V_{out} = -99.949V$$

Case C: Comparison — Ideal vs Practical EOA

Parameter	Case A: Ideal Op-Amp ($r_d = \infty, r_o = 0$)	Case B: Practical Op-Amp ($r_d = 2M\Omega, r_o = 75\Omega$)
Input Resistance (r_d)	∞ (infinite)	2 M Ω
Output Resistance (r_o)	0 (zero)	75 Ω
Current into Input Stage	0 (no loading on R_1)	Negligible — $r_d \gg R_1$
Voltage Drop at Output	0 (no drop across r_o)	Tiny — $r_o \ll R_2$
Output Voltage (V_{out})	-99.949 V	-99.949 V
Closed-Loop Gain	-100.000 (exact = $-R_2/R_1$)	-99.949
Deviation from Ideal	0 V / 0%	0.051 V / 0.051%
Approximation Accuracy	Reference (exact)	Extremely accurate
Condition for Accuracy	Always (by definition)	$r_d \gg R_1$ and $r_o \ll R_2$ must hold
Practical Impact on Gain	None (ideal reference)	Negligible for R_1, R_2 in k Ω –100k Ω range

Conclusion:

Case A (ideal: $r_d = \infty, r_o = 0$) gives $V_{out} = -100V$ exactly, corresponding to a closed-loop gain of exactly -100 ($= -R_2/R_1$). Case B (practical: $r_d = 2M\Omega, r_o = 75\Omega$) gives $V_{out} = -99.949V$, a closed-loop gain of -99.949 . The difference is only 0.051V (0.051%), confirming that the ideal op-amp approximation is extremely accurate here because (i) $r_d \gg R_1$, so negligible current is diverted into the input resistance, and (ii) $r_o \ll R_2$, so the output resistance introduces only a tiny voltage drop. In practice, the finite r_d and r_o of a real op-amp have almost no impact on the inverting amplifier gain when R_1 and R_2 are in the k Ω -to-100k Ω range, as long as $r_d \gg R_1$ and $r_o \ll R_2$ are satisfied.

Case D: LT Spice VCVS Model — Simulation

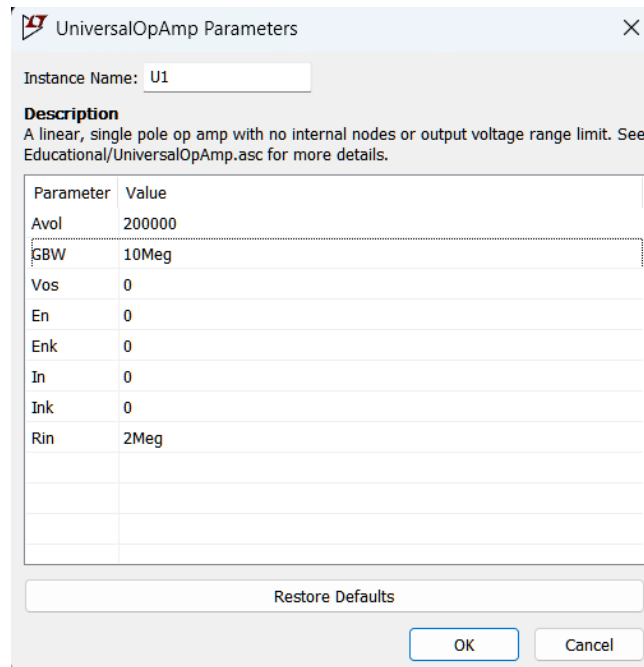


Fig 4: UniversalOpAmp parameters - $A_{vol} = 200000$, $R_{in} = 2M\Omega$

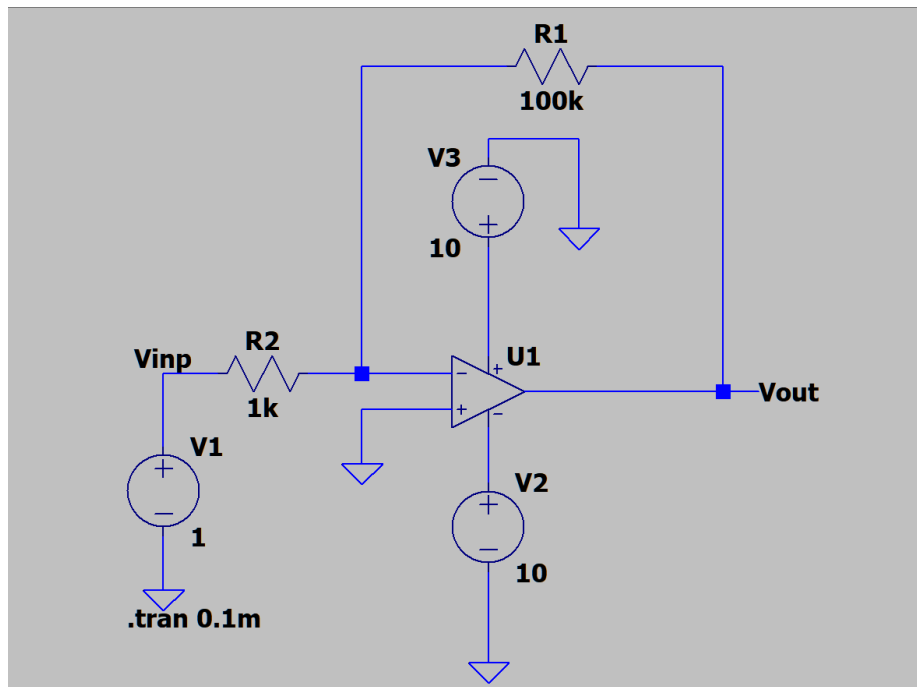


Fig 5: LT Spice circuit - Inverting amplifier with UniversalOpAmp ($R1 = 1k\Omega$, $Rf = 100k\Omega$, $\pm 10V$ supply), $V1 = 1V$ DC

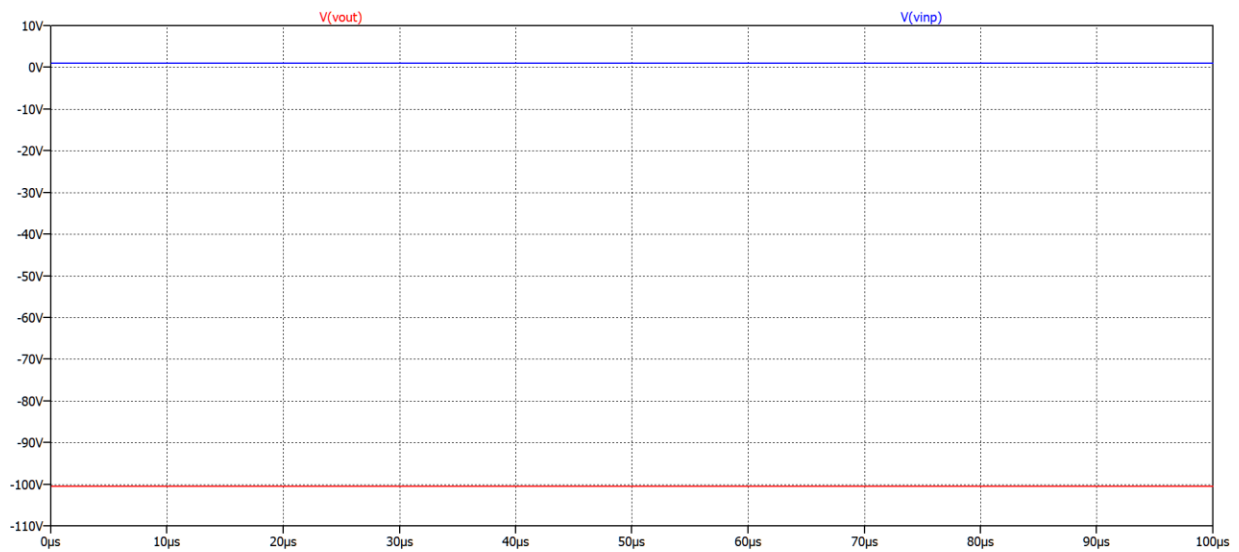


Fig 6: Transient result - $V(vout) = -100V$ (red), $V(vinp) = 1V$ (blue), confirming $V_o/V_i = -100$ for $V_i = 1V$

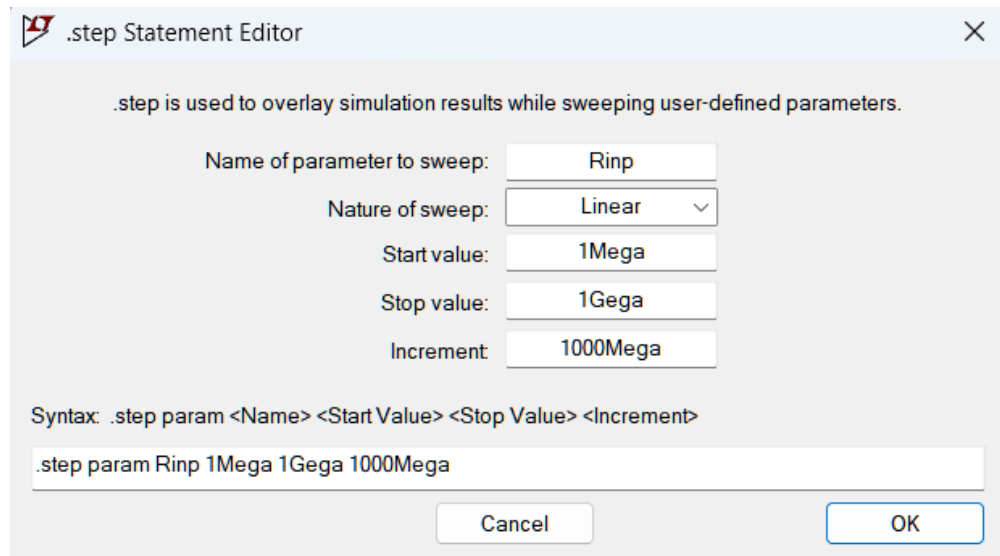


Fig 7: Sweep of R_{in} from $1M\Omega$ to $1G\Omega$ to observe effect of varying input resistance on V_{out}

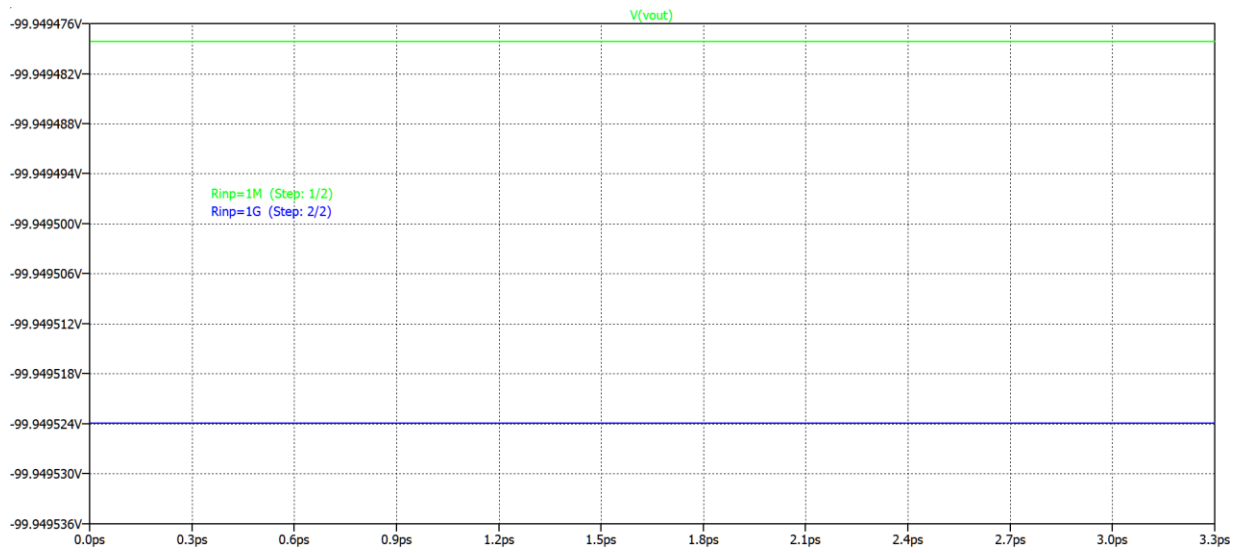


Fig 8: $V(vout)$ vs R_{inp} sweep - $R_{inp} = 1M$ gives -99.9495V (green), $R_{inp} = 1G$ gives -99.9495V (blue); negligible difference confirms $rd \gg R1$

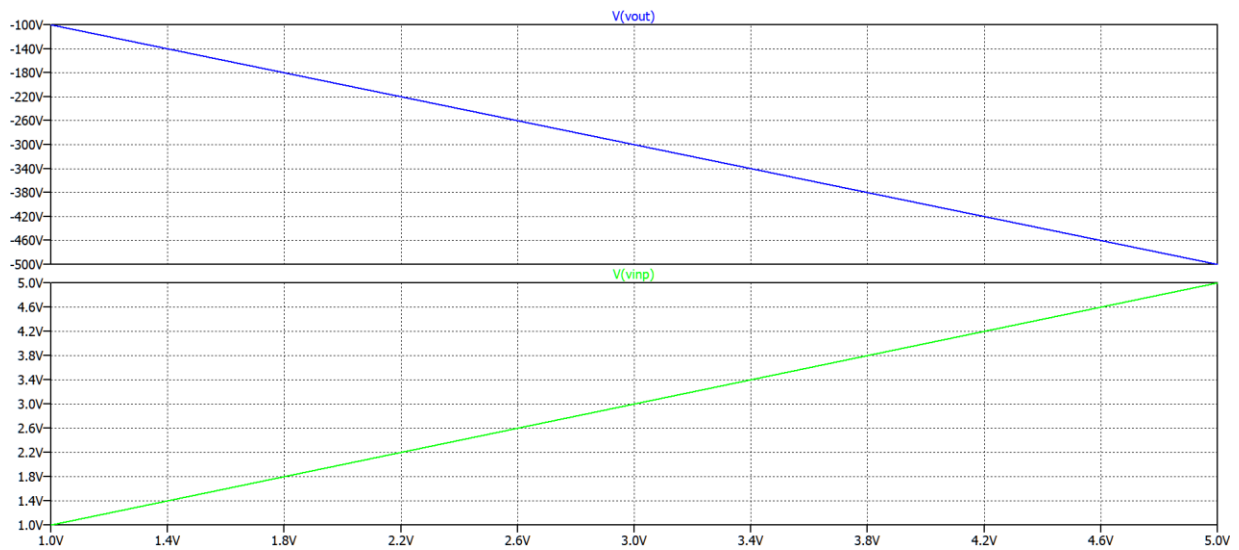


Fig 9: V_o/V_i plot sweeping V_{inp} from 1V to 5V - V_{out} scales linearly (top: -100V to -500V), confirming constant gain of -100 across input range

Problem 2: Loading effect and use of Buffer circuit

1. What is loading effect in the voltage divider network
2. Draw the two schematics in LT spice as show in the figures and compare the the output of the both
 - a. With different values of the R_L (from 1ohm to 100M Ohm), $R_s = 100\text{ohm}$ (fixed)
 - b. In case b sweep the gain of the buffer form 100 to 10k, and meaure the output (V_L)
 $R_s = 100\text{ ohm}$, $V_s = 1\text{V DC}$.
 - c. Do the same scenario as b but change the V_s to ac source (1kHz frquency)
 - d. Model the EOA using VCVS in LT Spice and plot the V_O/V_i by varying the input signal from 1V, to 5 V and then tweak the value of r_d to oberserve the changes in V_O/V_i

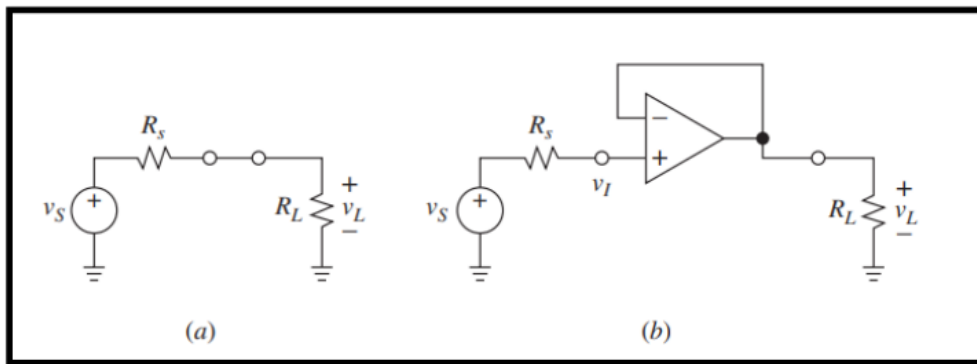


Fig 10: Problem 2 - (a) Voltage divider without buffer (loading effect),
(b) Voltage divider with op-amp buffer

Solution :-

1. Loading Effect in a Voltage Divider Network

When a load (R_L) is directly connected to the output of a voltage divider, it alters the effective resistance of the lower arm. This changes the output voltage from its unloaded (Thevenin) value. Specifically, R_L appears in parallel with the lower divider resistor, reducing the effective resistance and therefore reducing the output voltage — this is the loading effect. The severity of loading is captured by the ratio R_{th}/R_L : when $R_{th}/R_L \ll 1$ ($R_L \gg R_{th}$), loading is negligible. When R_L approaches R_{th} , the output voltage drops significantly.

2(a): Without Buffer — RL Swept from $1k\Omega$ to $100k\Omega$

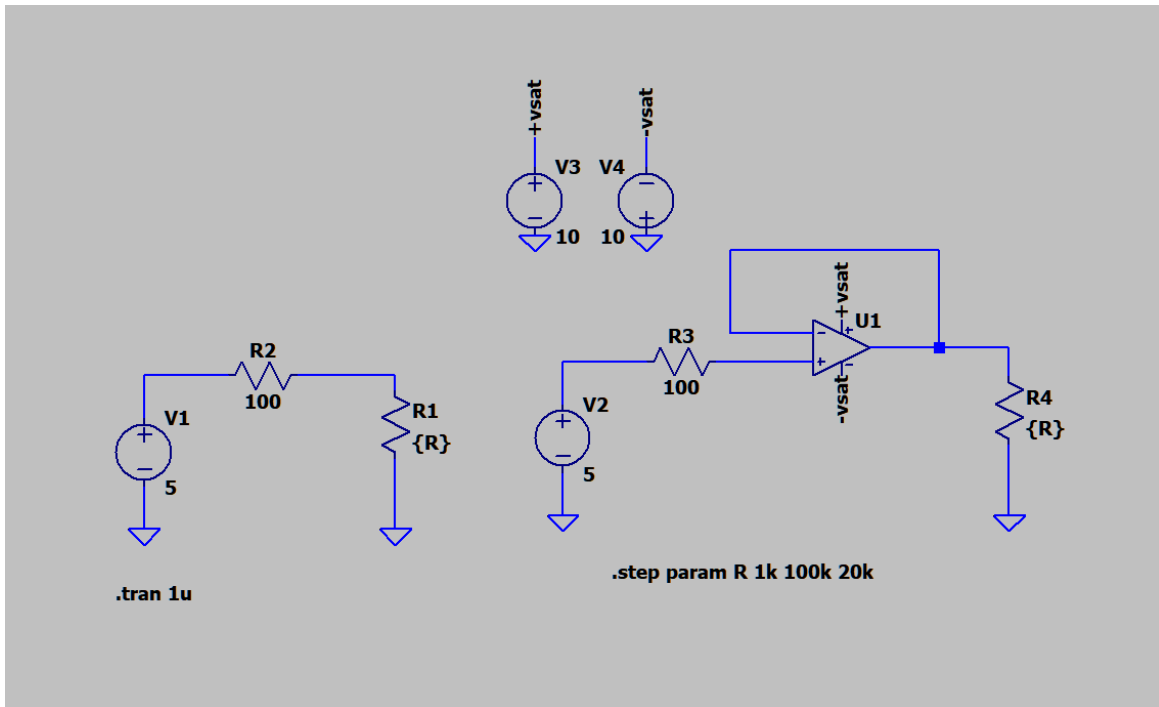


Fig 11: LT Spice schematic - Left: direct connection (a) with $R_s = 100\Omega$, $R_L = \{R\}$ swept; Right: buffer circuit (b) with $R_s = 100\Omega$, A_{ol} swept, $R_L = \{R\}$

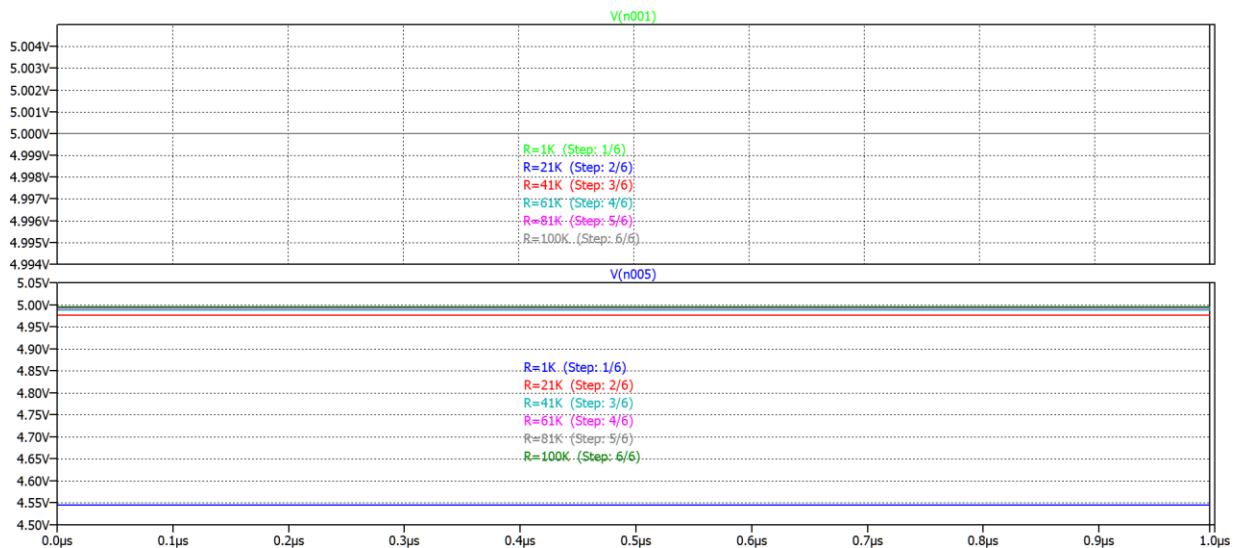


Fig 12: RL parametric sweep results ($R = 1k, 21k, 41k, 61k, 81k, 100k$). Top: $V(n001)$ without buffer - voltage drops noticeably at low R_L . Bottom: $V(n005)$ with buffer - output remains nearly constant ($\sim 5V$) regardless of R_L

Observation :-

Without a buffer (circuit a), the output voltage decreases as R_L decreases, since R_L loads the divider. At $R = 1k$ the output is noticeably lower than 5V. With the op-amp buffer (circuit b), the output stays close to 5V for all R_L values, demonstrating that the buffer isolates the divider from the load and eliminates the loading effect.

2(b): With Buffer — Aol Swept (DC, $R_s = 100\Omega$, $V_s = 1V$)

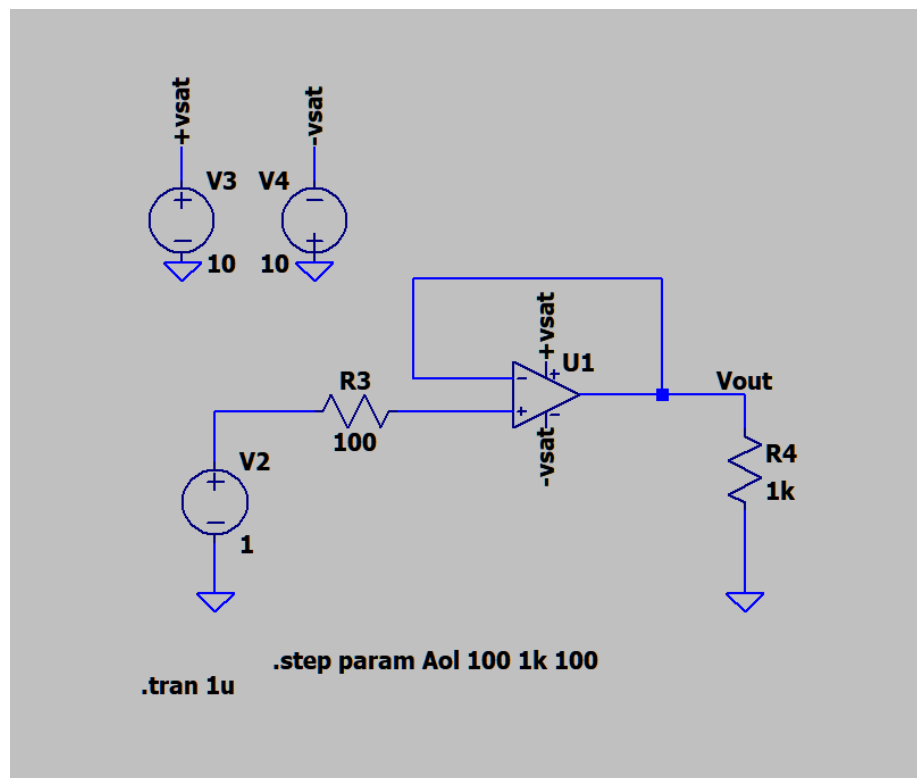


Fig 12: Buffer circuit with .step param Aol from 100 to 1k in steps of 100, $V_2 = 1V$ DC

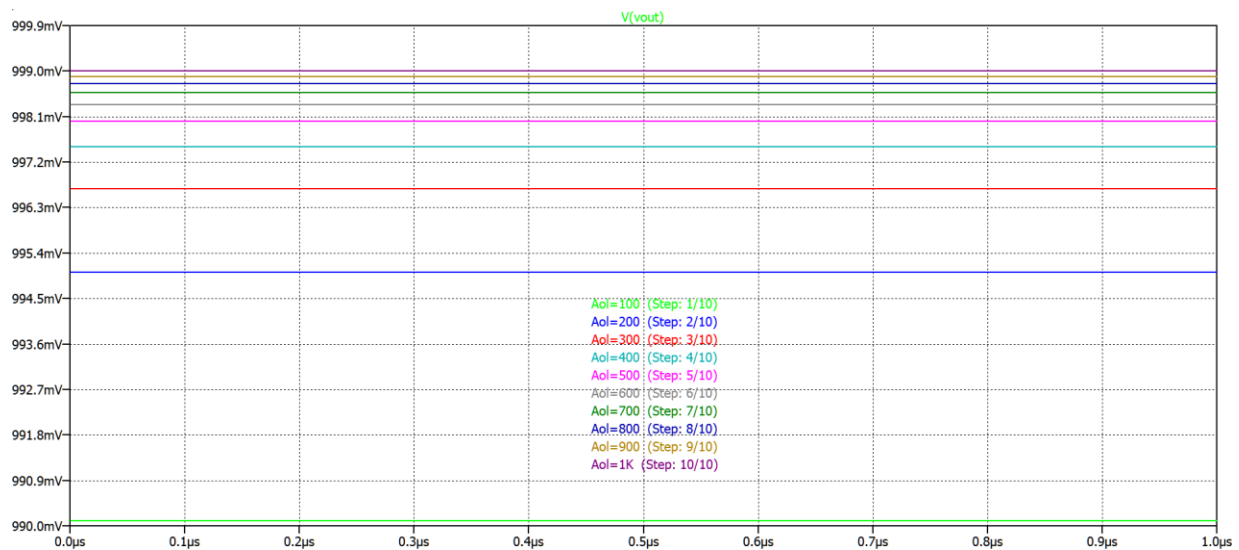


Fig 13: $V(vout)$ for Aol sweep (100 to 1K) - as Aol increases, $Vout$ converges toward 1V; lower Aol gives slight deviation

Observation :-

As Aol increases from 100 to 1000, the output voltage at $V(vout)$ approaches 1V more closely. For Aol = 100 the output is about 994.5mV; for Aol = 1000 it reaches approximately 999mV. This shows that a higher open-loop gain makes the buffer track the input more accurately, reinforcing why practical op-amps ($Aol = 10^5$ to 10^6) are effectively ideal for this application.

2(c): With Buffer — Aol Swept (AC 1kHz, $R_s = 100\Omega$, $V_s = 5V$)

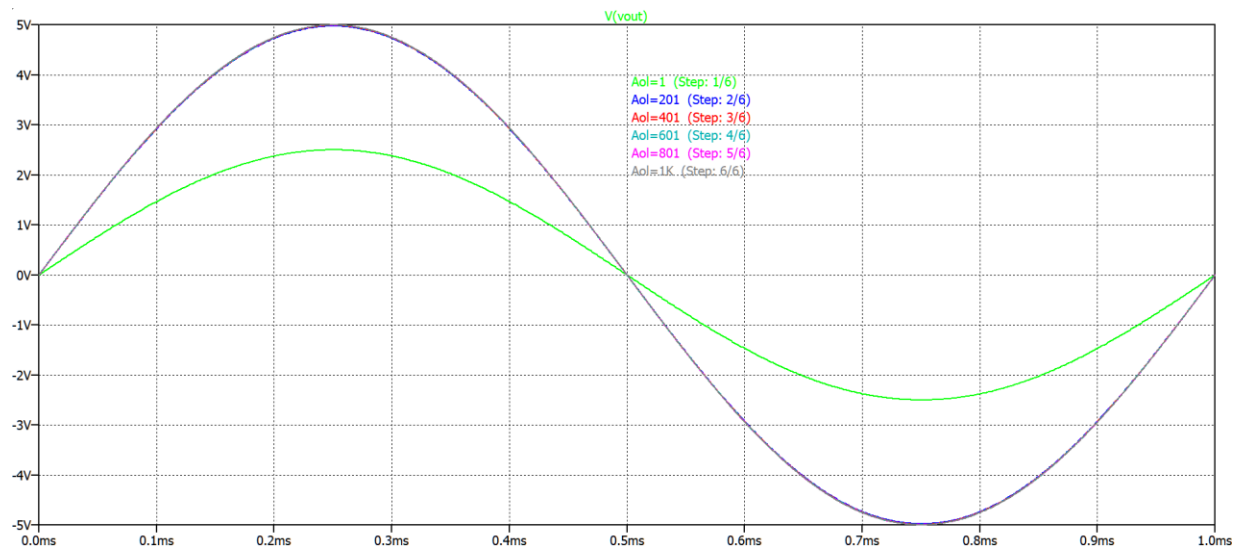


Fig 14: AC transient output $V(vout)$ with Aol swept (1, 201, 401, 601, 801, 1k) - for Aol = 1 the output amplitude is reduced (~2.5V peak) while higher Aol values produce clean unity-gain sine waves at 5V peak

Observation :-

For very low A_{ol} ($A_{ol} = 1$), the feedback loop is insufficient to maintain unity gain and the output sine wave is attenuated to roughly half the input amplitude. As A_{ol} increases, the output amplitude approaches the input amplitude. For $A_{ol} \geq 401$ the output is virtually indistinguishable from the input, confirming that a sufficiently high open-loop gain is required for accurate buffering even at AC frequencies.

Problem 3: Op-config using real Op-Amp in LT spice

1. Build an inverting amplifier circuit with an amplification of 100 using a LM741 op amp. Do the ac analysis and show the input and output voltage plot.
2. Build an inverting amplifier circuit with an amplification of 100 using a LT1055 op amp. Do the ac analysis and show the input and output voltage plot.
3. Describe the difference between above cases? Hint: Check the i/p pairs!

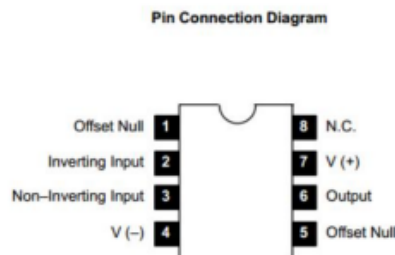


Fig 15: Problem 3 - LM741 pin connection diagram (8-pin DIP: Offset Null, Inverting Input, Non-Inverting Input, V(-), Offset Null, Output, V(+), N.C.)

Hand Calculations :-

Design specification derivation and practical considerations using the LM741 datasheet as the reference:

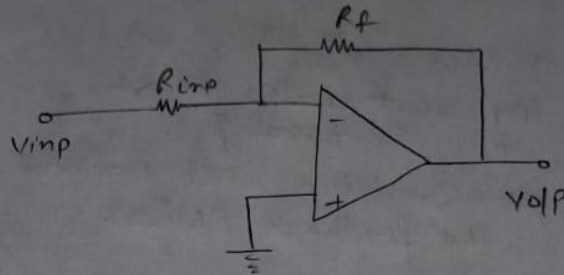
6.5 Electrical Characteristics, LM741⁽¹⁾

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
Input offset voltage		$R_S \leq 10\text{ k}\Omega$	$T_A = 25^\circ\text{C}$		1	5	mV
			$T_{AMIN} \leq T_A \leq T_{AMAX}$			6	mV
Input offset voltage adjustment range		$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{ V}$			± 15		mV
Input offset current		$T_A = 25^\circ\text{C}$			20	200	nA
		$T_{AMIN} \leq T_A \leq T_{AMAX}$			85	500	
Input bias current		$T_A = 25^\circ\text{C}$			80	500	nA
		$T_{AMIN} \leq T_A \leq T_{AMAX}$				1.5	μA
Input resistance		$T_A = 25^\circ\text{C}$, $V_S = \pm 20\text{ V}$			0.3	2	M Ω
Input voltage range		$T_{AMIN} \leq T_A \leq T_{AMAX}$			± 12	± 13	V
Large signal voltage gain		$V_S = \pm 15\text{ V}$, $V_O = \pm 10\text{ V}$, $R_L \geq 2\text{ k}\Omega$	$T_A = 25^\circ\text{C}$	50	200	V/mV	
			$T_{AMIN} \leq T_A \leq T_{AMAX}$		25		
Output voltage swing		$V_S = \pm 15\text{ V}$	$R_L \geq 10\text{ k}\Omega$	± 12	± 14	V	
			$R_L \geq 2\text{ k}\Omega$	± 10	± 13		
Output short circuit current		$T_A = 25^\circ\text{C}$			25	mA	
Common-mode rejection ratio		$R_S \leq 10\text{ }\Omega$, $V_{CM} = \pm 12\text{ V}$, $T_{AMIN} \leq T_A \leq T_{AMAX}$			80	95	dB
Supply voltage rejection ratio		$V_S = \pm 20\text{ V}$ to $V_S = \pm 5\text{ V}$, $R_S \leq 10\text{ }\Omega$, $T_{AMIN} \leq T_A \leq T_{AMAX}$			86	96	dB
Transient response	Rise time	$T_A = 25^\circ\text{C}$, unity gain			0.3	μs	
	Overshoot				5%		
Slew rate		$T_A = 25^\circ\text{C}$, unity gain			0.5	V/ μs	
Supply current		$T_A = 25^\circ\text{C}$			1.7	2.8	mA
Power consumption		$V_S = \pm 15\text{ V}$	$T_A = 25^\circ\text{C}$	50	85	mW	
			$T_A = T_{AMIN}$	60	100		
			$T_A = T_{AMAX}$	45	75		

(1) Unless otherwise specified, these specifications apply for $V_S = \pm 15\text{ V}$, $-55^\circ\text{C} \leq T_A \leq +125^\circ\text{C}$ (LM741/LM741A). For the LM741C/LM741E, these specifications are limited to $0^\circ\text{C} \leq T_A \leq +70^\circ\text{C}$.

Fig 16: LM741 Electrical Characteristics (Texas Instruments) - key parameters: $A_{ol} = 50\text{-}200\text{ V/mV}$, $R_{in} = 0.3\text{-}2\text{ M}\Omega$, $V_o\text{ swing} = \pm 10\text{ V}$ at $R_L \geq 2\text{ k}\Omega$, $SR = 0.5\text{ V}/\mu\text{s}$, $GBW = 1\text{ MHz}$

HAND CALCULATIONS



$$A = 100$$

Layer 0: Design Specification.

- ① $V_S = \pm 15$ → select as design parameter from datasheet.
- ② Load = not stated.
- ③ z/p amplitude = not stated → IC's o/p swing limit will tell us about max amplitude
- ④ z/p frequency = Not stated
IC → GBW & slew rate will tell.
- ⑤ DC Accuracy ⇒ Not stated

Why?

Beoz every calculation I will do is a ratio of Design value over limit, limit comes from Datasheet, The Design value comes from requirement.

Layer 1: Datasheet. (Static Analysis).

- ① z/p offset voltage: at 25°C max 6mV.

$$\text{ie } V_{\text{off offset}} = \underset{\substack{\uparrow \\ \text{gain}}}{6\text{mV}} \times 100 = 600\text{mV DC error at o/p}$$

observation:

It can't be eliminated by the resistors
beoz its the property of IC (Manufacturing defect in Trans)
So (later Handle) That's large

Fig 17: Hand calculation Layer 0 - Design specification: $V_S = \pm 15\text{V}$ from datasheet, gain $A = 100$, $V_{os} = 6\text{mV}$ max causes 600mV DC error at output

② Input bias current : MAX 500nA (25°C)

if R_f is larger (MΩ)

so, V_{out} is also large
i.e. error ↑

so, this constrain will bound
the feedback Resistor.

③ Input offset current : MAX 200nA (25°C)

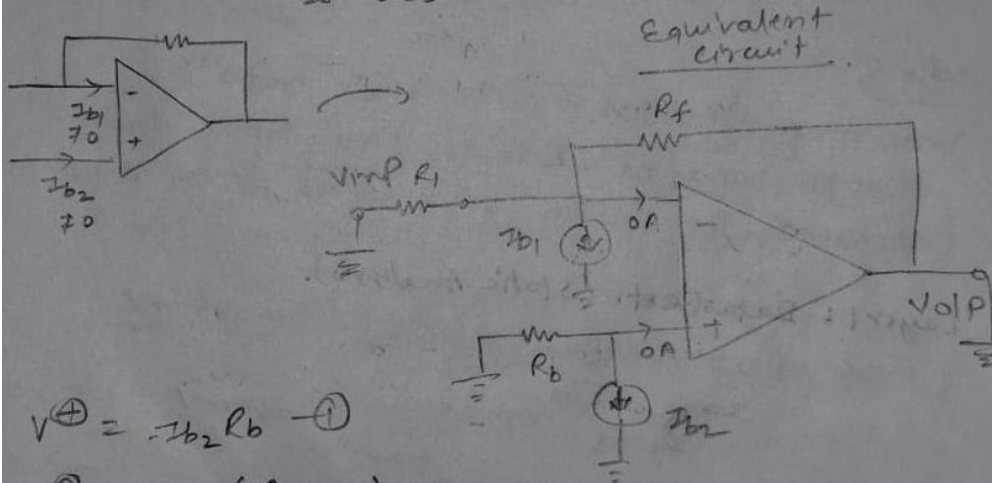
↳ An OpAmp I/P terminals are not ideal,
The differential pair transistors (BJT in LM741)
at the I/P need a small base current
to stay in their active region.

This current exists for the BJT to
maintain its operating region.

but due to Manufacturing
defect $\beta_1 \neq \beta_2$ (diff Pair).

so, $I_{b1} \neq I_{b2}$

so $I_{OS} = I_{b1} - I_{b2}$



$$V^{\oplus} = -I_{b2} R_b \quad \text{--- (1)}$$

$$V^{\ominus} = I_{b1} (R_f \parallel R_1) \quad \text{--- (2)}$$

$$V_{inP} \& V_{outP} = 0V$$

$\therefore V^{\ominus}$ is not occurring
bcz of V_{inP}

It was appearing
bcz of BJT base current

Fig 18: Hand calculation Layer 1 (Static Analysis) - Input bias current (max 500nA), input offset current effects, and $R_b = R_f \parallel R_1$ technique to minimize offset error

So for $V_{O1} = 0$ (offset of opamp)

$$V_{O1} = A_{OL} \times V_d$$

$$0 = A_{OL} \times (V^{\oplus} - V^{\ominus})$$

$$V^{\oplus} = V^{\ominus}$$

$$-I_{b1} (R_f \parallel R_1) = -I_{b2} R_b \quad \text{--- (A)}$$

but $I_{b1} \neq I_{b2}$ (never).

$$I_{b1} - I_{b2} = I_{OS} = \Delta I$$

$$\text{Now, } I_{b1} = I_b + \frac{\Delta I}{2} \quad \text{--- (1)}$$

$$I_{b2} = I_b - \frac{\Delta I}{2} \quad \text{--- (2)}$$

$$V^{\ominus} = +I_{b1} \times (R_f \parallel R_1) = \left(I_b + \frac{\Delta I}{2}\right) (R_f \parallel R_1)$$

$$V^{\oplus} = -I_{b2} \times R_b = -\left(I_b - \frac{I_{OS}}{2}\right) \times R_b$$

Now, if $R_b = R_f \parallel R_1$

then due to eq (A) $I_{b1} \neq I_{b2}$
ie $V_{O1} \neq 0$ can't
offset

but we can reduce it upto some extent.

if $R_b = R_f \parallel R_1$

$$\text{So } V^+ = I_{b2} R_b, \quad V^{\ominus} = I_{b1} R_b$$

$$\text{ie } \Delta V = V^{\oplus} - V^{\ominus}$$

$$\Delta V = R_b (I_{b2} - I_{b1}) = R_b I_{OS}$$

→ always there at I/P pins.

ie

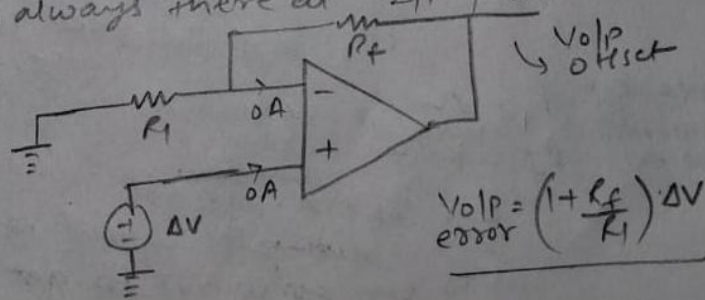


Fig 19: Hand calculation Layer 1 cont. - Derivation of output offset due to I_{OS} : op error = $I_{OS} \times R_f$, with $I_{OS} = 200\text{nA}$ (datasheet). Shows $R_b = R_1 \parallel R_f$ for compensation

$$o/p \text{ error} = (Z_{os} \times R_B) \times \left(1 + \frac{R_f}{R_i}\right)$$

$$R_B = R_i \parallel R_f$$

$$= Z_{os} \times \frac{R_i R_f}{R_i + R_f} \times \frac{R_i + R_f}{R_i}$$

$$\boxed{o/p \text{ error} = Z_{os} R_f}$$

$Z_{os} = 200 \text{ nA}$
at 25°C
(datasheet).

So if $R_f \uparrow$ o/p error $\uparrow$ also.

④ o/p voltage swing :- $\pm 10 \text{ V}$ at $R_L \geq 2 \text{ k}\Omega$
 $V_{CC} = \pm 15 \text{ V}$

So o/p is bounded
in b/w $\pm 10 \text{ V}$.

⑤ z/p voltage range :- $\pm 12 \text{ V}$ at 25°C

⑥ slew rate = $0.5 \text{ V}/\mu\text{s}$

⑦ GBW = 1 MHz (not in datasheet
from simulation).

Layer 2:- Resistances values,

$$\left|\frac{R_f}{R_i}\right| = 100$$

So, $R_f \propto R_i$

if $R_f \rightarrow \text{M}\Omega$
 $R_i \rightarrow \text{k}\Omega$

o/p error significant

$$\leq 106 \times 10^9$$

$$\leq \text{mV}$$

if $R_i = \Omega$
 $R_f = \text{k}\Omega$

$\rightarrow$ Ib error small

but z/p current is too high

it can force BJT to get off from their
Active region

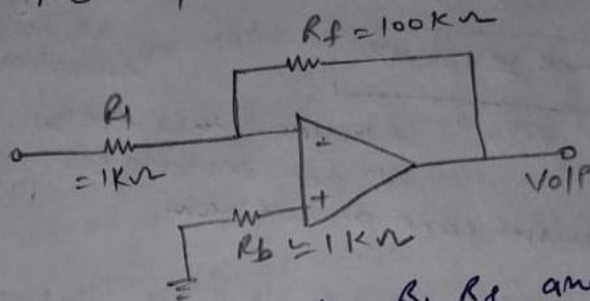
Fig 20: Hand calculation Layer 2 - Resistance values: $|R_f/R_i| = 100$, so $R_f = 100 \text{ k}\Omega$, $R_i = 1 \text{ k}\Omega$ (safe range). If $R_f \rightarrow \text{M}\Omega$, Ib error becomes large. If $R_i \rightarrow 1 \Omega$, BJT may leave active region

so safe (intuitive) range
 $R_1 = 1k\Omega$
 $R_f = 100k\Omega$

Now due to input offset current $\rightarrow$ Voltage error $= I_{os} R_f$
 $= 200nA \times R_f$

but due to input offset voltage $= 6mV$ ie $\rightarrow$ Voltage error $= 6mV \times 100$
 (fixed can't in our hands) $= 600mV$ fixed.

So, $R_b = R_1 \parallel R_f = 990\Omega \approx 1k\Omega$



Conclusion:- why we do R_1, R_f analysis first not V_{in} & frequency one?

Beoz:- first we always perform static analysis $\therefore$ freq is dynamic, so we always set the first analysis when time is not changing ie DC operating points, ie ckt is alive but nothing is moving.

so, in static analysis

R_f, R_b, R_1

$V_{out} (max)$, $V_{in} (max)$
 PK-PK

} all are known to us.

Fig 21: Hand calculation Layer 2 cont. - Safe range $R_1 = 1k\Omega$, $R_f = 100k\Omega$. Due to input offset voltage (6mV fixed), voltage error $= 6mV \times 100 = 600mV$ (fixed, cannot be eliminated by resistors). $R_b = R_1 \parallel R_f = 990\Omega \approx 1k\Omega$

Layer 3: Dynamic Analysis:

$$f_{3dB} = \frac{GB}{|A_v|}$$

$$|A_v| = \frac{R_f}{R_i}$$

$$GBW = 1\text{MHz} = 100 \times BW$$

$$BW = 10\text{kHz}$$

$$f_{3dB} = 10\text{kHz}$$

doesn't care abt Amplitude

After that freq
VOP starts
attenuating

slew rate: $0.5\text{V}/\mu\text{s}$

$$f_{max} = \frac{0.5 \times 10^6}{2\pi \times 5}$$

VOP
Peak = 5V
(we want)
let say
as design
specification

$$f_{max} \approx 16\text{kHz}$$

ie beyond this freq
O/P got distorted.
doesn't care about gain

* If VOP-peak is small $\rightarrow f_{SR}$ is large

$\downarrow$
SR Not a
Problem

* If VOP-peak is large $\rightarrow f_{SR}$ is small

$\downarrow$
SR becomes
Problem

ie at some amplitude, f_{SR} will equal f_{GB}
That's crossover point

Below that amplitude GBW governs

Above it, SR governs.

$$f_{GBW} = f_{SR}$$

$$\frac{GBW}{|A|} = \frac{SR}{2\pi \times V_{out_cross\ over}}$$

Fig 22: Hand calculation Layer 3 (Dynamic Analysis) - $GBW = 1\text{MHz}$, $|A_v| = 100 \rightarrow f_{3dB} = 10\text{kHz}$, $f_{max}(SR) = SR/(2\pi \times V_{out_peak}) = 0.5 \times 10^6/(2\pi \times 5) \approx 16\text{kHz}$

$$V_{out_cross\ over} = \frac{SR \times |A_v|}{2\pi \times GBW} = \frac{0.5 \times 10^6 \times 100}{2\pi \times 10^6}$$

$$V_{out} = 7.96V \text{ (peak)}$$

So, It's a tradeoff b/w GBW & slew rate but SR is dominating becoz it changes the signal shape & for designing of amplifier we have to take care of it (High priority)

∴ for 5V inp signal $f_{max} \leq 16kHz$
 $f_{3d} = 10kHz$
 so,
 we can take 1kHz → Best here

also, for 10V inp signal $f_{max} \leq 8kHz$
 i.e. Here strictly we have to follow SR
 no choice so, again 1kHz
 best choice.

1kHz
 It can handle both small & large signals.

layer 4: Power = $2.8mA \times 30V$ (datasheet)
 = 84mW

Thermal $\uparrow$ note = $84mW \times 150^\circ C/W = 12.6^\circ C$

i.e. $25^\circ C + 12.6^\circ = 37.6^\circ C < 150^\circ C$
PASS.

o/p current = $\frac{2.5V}{R_b} = \frac{2.5}{100} = 0.025mA < 25mA$
PASS.

Fig 23: Hand calculation Layer 3 cont. - Crossover analysis: GBW vs SR limits. For 5V peak, $f_{SR} = 16kHz > f_{GBW} = 10kHz \rightarrow GBW$ governs. For 10V peak, $f_{SR} = 8kHz < f_{GBW} = 10kHz \rightarrow SR$ governs. Best choice: 1kHz signal

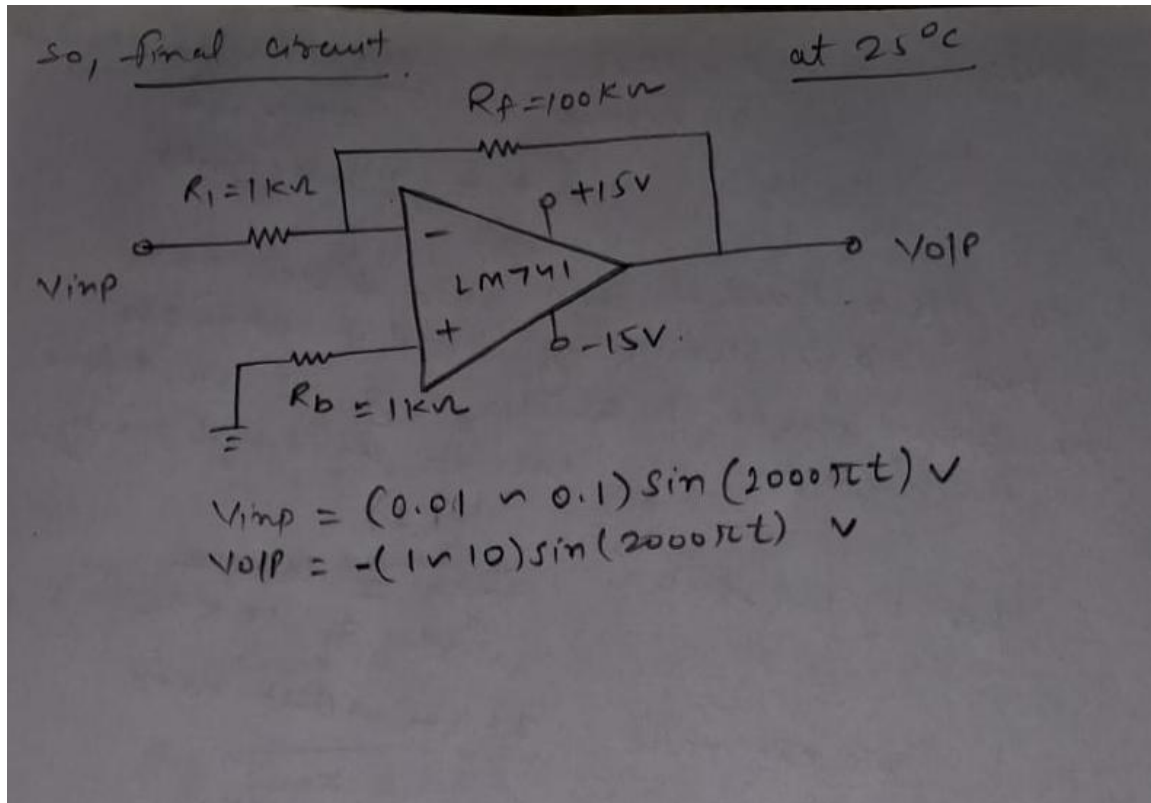


Fig 24: Final LM741 inverting amplifier design: $R_1 = 1k\Omega$, $R_f = 100k\Omega$, $R_b = 1k\Omega$, $V_s = \pm 15V$.
 $V_{inp} = (0.01 \text{ to } 0.1) \sin(2000\pi t) \text{ V}$, $V_{out} = -(1 \text{ to } 10) \sin(2000\pi t) \text{ V}$

LM741 Inverting Amplifier (Gain = 100) — Simulation

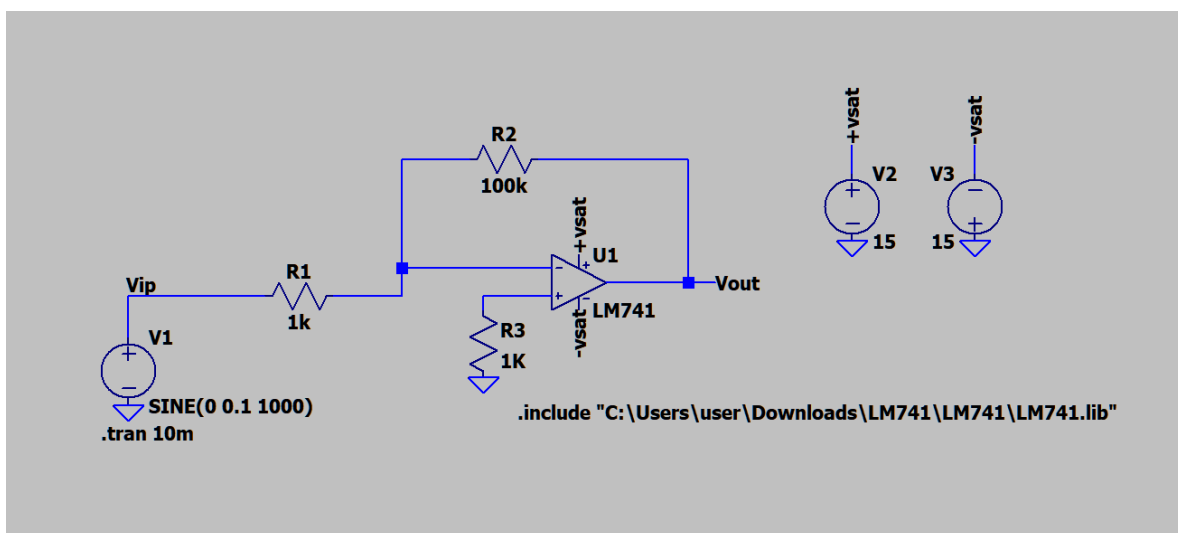


Fig 25: LM741 LT Spice circuit - $R_1 = 1k\Omega$, $R_2 = 100k\Omega$, $R_3 = 1k\Omega$ (R_b), $V_1 = \text{SINE}(0, 0.1, 1000)$, $\pm 15V$ supply, $.tran 10m$

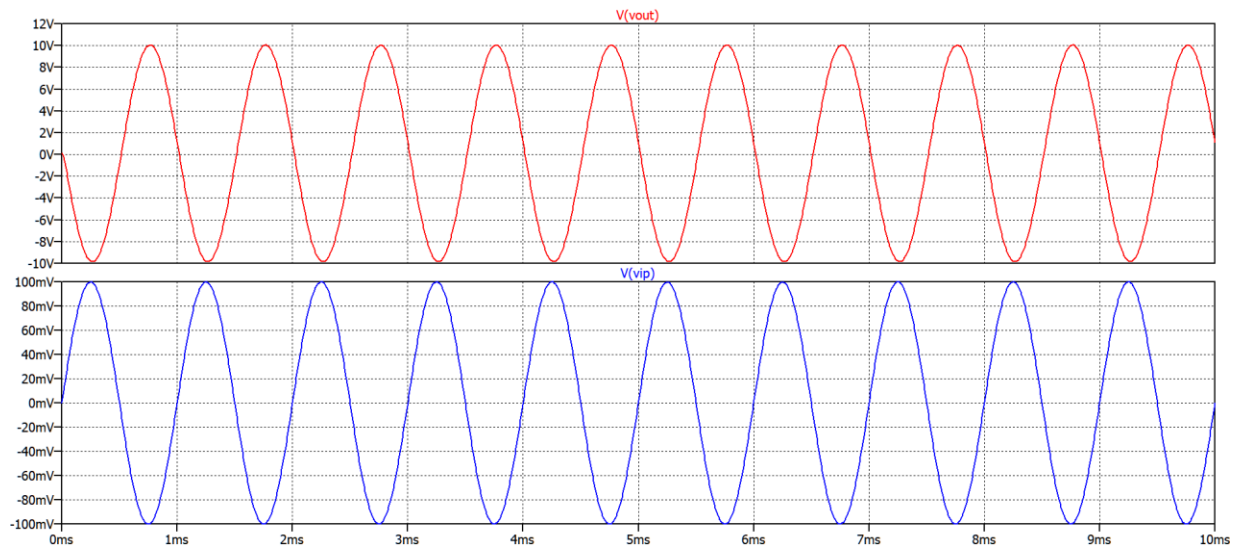


Fig 26: LM741 transient - $V(vout)$ peak = $\pm 10V$ (red), $V(vip)$ peak = $\pm 100mV$ (blue). Gain = $10/0.1 = 100$. Clean inversion confirmed

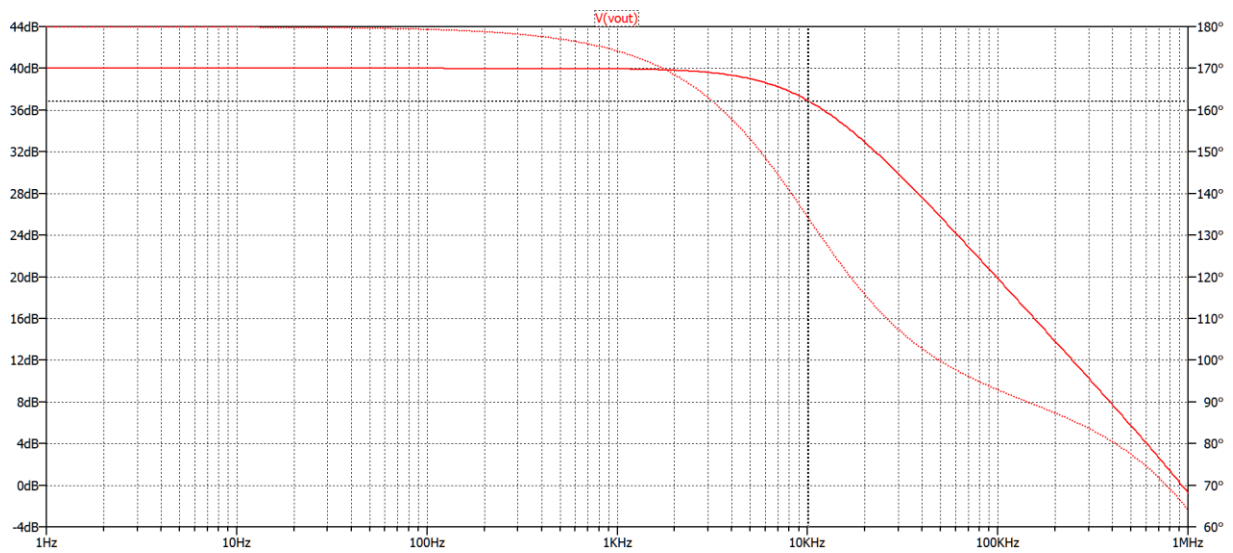


Fig 27: LM741 AC analysis (Bode plot) - flat gain $\approx 40dB$ from 1Hz to $\sim 10kHz$, then rolls off at -20dB/dec. Phase starts near 180° and drops

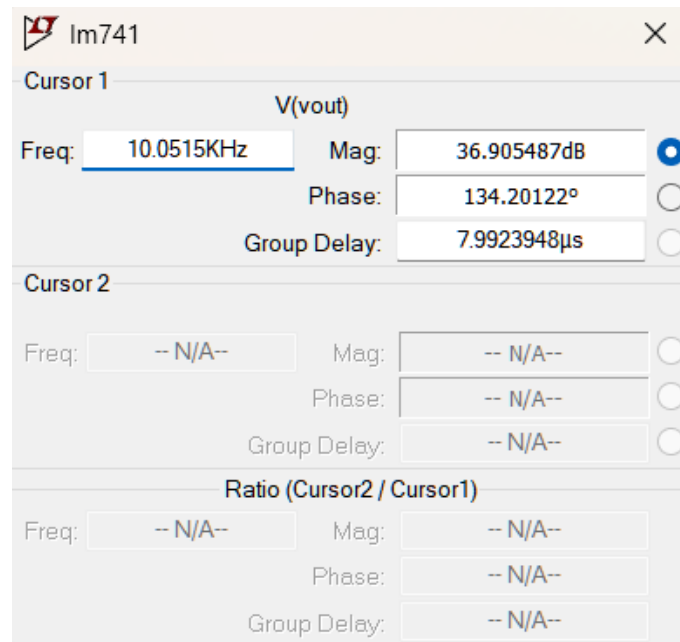


Fig 28: LM741 Bode cursor measurement at -3dB point: Freq = 10.05kHz, Magnitude = 36.91dB ($\approx 40 - 3\text{dB}$), confirming $f_{3\text{dB}} \approx 10\text{kHz}$ as predicted by $GBW/|A_v| = 1\text{MHz}/100$

2. LT1055 Inverting Amplifier (Gain = 100)

LT1055/LT1056

ELECTRICAL CHARACTERISTICS

$T_A = 25^\circ\text{C}$. $V_S = \pm 15\text{V}$, $V_{CM} = 0\text{V}$ unless otherwise noted.

SYMBOL	PARAMETER	CONDITIONS	LT1055CS8/LT1056CS8			UNITS
			MIN	TYP	MAX	
V_{OS}	Input Offset Voltage (Note 2)			500	1500	μV
I_{OS}	Input Offset Current	Fully Warm Up		5	30	pA
I_B	Input Bias Current	Fully Warm Up $V_{CM} = 10\text{V}$		± 30 30	± 100 150	pA
	Input Resistance	Differential Common Mode $V_{CM} = -11\text{V}$ to 8V $V_{CM} = 8\text{V}$ to 11V		0.4 0.4 0.05		$\text{T}\Omega$ $\text{T}\Omega$ $\text{T}\Omega$
	Input Capacitance			4		pF
e_n	Input Noise Voltage	0.1Hz to 10Hz LT1055 LT1056		2.5 3.5		μV_{P-P} μV_{P-P}
	Input Noise Voltage Density	$f_0 = 10\text{Hz}$ (Note 4) $f_0 = 1\text{kHz}$ (Note 4)		35 15	70 22	$\text{nV}/\sqrt{\text{Hz}}$ $\text{nV}/\sqrt{\text{Hz}}$
i_n	Input Noise Current Density	$f_0 = 10\text{Hz}$, 1kHz (Note 5)		2.5	10	$\text{fA}/\sqrt{\text{Hz}}$
A_{VOL}	Large-Signal Voltage Gain	$V_0 = \pm 10\text{V}$ $R_L = 2\text{k}$ $R_L = 1\text{k}$	120 100	400 300		V/mV V/mV
	Input Voltage Range		± 11	± 12		V
CMRR	Common Mode Rejection Ratio	$V_{CM} = \pm 11\text{V}$	83	98		dB
PSRR	Power Supply Rejection Ratio	$V_S = \pm 10\text{V}$ to $\pm 18\text{V}$	88	104		dB
V_{OUT}	Output Voltage Swing	$R_L = 2\text{K}$	± 12	± 13.2		V
SR	Slew Rate	LT1055	7.5	12		$\text{V}/\mu\text{s}$
		LT1056	9.0	14		$\text{V}/\mu\text{s}$
GBW	Gain Bandwidth Product	$f = 1\text{MHz}$ LT1055		4.5		MHz
		LT1056		5.5		MHz
I_S	Supply Current	LT1055	2.8	4.0		mA
		LT1056	5.0	7.0		mA
	Offset Voltage Adjustment Range	$R_{POT} = 100\text{k}$		± 5		mV

The ● denotes the specifications which apply over the temperature range $0^\circ\text{C} \leq T_A \leq 70^\circ\text{C}$. $V_S = \pm 15\text{V}$, $V_{CM} = 0\text{V}$ unless otherwise noted.

SYMBOL	PARAMETER	CONDITIONS		LT1055CS8/LT1056CS8			UNITS
				MIN	TYP	MAX	
V_{OS}	Input Offset Voltage (Note 2)		●		800	2200	μV
	Average Temperature Coefficient of Input Offset Voltage		●		4	15	$\mu\text{V}/^\circ\text{C}$
I_{OS}	Input Offset Current	Warm Up, $T_A = 70^\circ\text{C}$	●		18	150	pA
I_B	Input Bias Current	Warm Up, $T_A = 70^\circ\text{C}$	●		± 60	± 400	pA
A_{VOL}	Large-Signal Voltage Gain	$V_0 = \pm 10\text{V}$, $R_L = 2\text{k}$	●	60	250		V/mV
CMRR	Common Mode Rejection Ratio	$V_{CM} = \pm 10.5\text{V}$	●	82	98		dB
PSRR	Power Supply Rejection Ratio	$V_S = \pm 10\text{V}$ to $\pm 18\text{V}$	●	87	103		dB
V_{OUT}	Output Voltage Swing	$R_L = 2\text{K}$	●	± 12	± 13.1		V

Fig 29: LT1055 Electrical Characteristics - key differences vs LM741: $V_{OS} = 500\mu\text{V}$ (max) vs 6mV , $I_B = \pm 30\text{pA}$ vs 80nA , $\text{GBW} = 4.5\text{MHz}$ vs 1MHz , $\text{SR} = 7.5 \text{ V}/\mu\text{s}$ vs $0.5 \text{ V}/\mu\text{s}$

Hand Calculations (LT1055) :-

Layer 0: Design Specification

Same reasoning as LM741. Gain $A = 100$. Design unknowns are resolved from the datasheet:

- ① $V_s = \pm 15V \rightarrow$ selected as design parameter from datasheet.
- ② Load = not stated.
- ③ i/p amplitude = not stated $\rightarrow$ IC's o/p swing limit will tell us about max amplitude.
- ④ i/p frequency = not stated $\rightarrow$ IC's GBW & slew rate will tell.
- ⑤ DC Accuracy = not stated.

Why? Every calculation is a ratio of design value over limit. Limit comes from datasheet; design value comes from requirement.

Layer 1: Datasheet — Static Analysis

① Input Offset Voltage

$V_{os} = 500 \mu V$ (max $1500 \mu V$) at $25^\circ C$ — from LT1055 datasheet.

ie $V_{olp\ offset} = V_{os} \times \text{Gain} = 500 \mu V \times 100 = 50 \text{ mV}$ DC error at o/p.

Observation: $V_{os} (\text{LT1055}) = 500 \mu V$ vs $V_{os} (\text{LM741}) = 6 \text{ mV} \rightarrow$ LT1055 is $12\times$ better.

Error = 50 mV (vs 600 mV in LM741). It still cannot be eliminated by resistors — it is a manufacturing property of the JFET diff pair (handle later with R_b).

② Input Bias Current

$I_b = \pm 30 \text{ pA}$ (max $\pm 100 \text{ pA}$) at $25^\circ C$ — LT1055 uses a JFET input diff pair.

If R_f is large ($M\Omega$) $\rightarrow$ V_{out} error $\uparrow$. But $I_b = 30 \text{ pA}$ is extremely small ($\sim 3000\times$ smaller than LM741's 80 nA), so this constraint is very relaxed. Still bounds R_f .

③ Input Offset Current

$I_{os} = 5 \text{ pA}$ (max 30 pA) at 25°C.

Unlike LM741 (BJT), LT1055 uses a JFET diff pair at the input. No BJT base current is needed, so I_b and I_{os} are $\sim 3000\times$ smaller. V_{inp} does NOT cause an offset at V_{olp} in the same way — the error from I_{os} is effectively negligible here. (It was appearing in LM741 because of BJT base current.)

For $V_{O1} = 0$ (output of op-amp in ideal analysis):

$$V_{O1} = A_{ol} \times V_d$$

$$0 = A_{ol} \times (V^+ - V^-) \rightarrow V^+ = V^-$$

$$-I_{b1}(R_f \parallel R_1) = I_{b2} \times R_b \text{ — (A)}$$

but $I_{b1} \approx I_{b2}$ (JFET: both $\sim 30 \text{ pA}$, nearly matched)

$$\therefore I_{b1} - I_{b2} = I_{os} = \Delta I \approx 5 \text{ pA (very small)}$$

$$\text{Now, } I_{b1} = I_b + \Delta I/2 \text{ — (1)}$$

$$I_{b2} = I_b - \Delta I/2 \text{ — (2)}$$

$$V^- = +I_{b1} \times (R_f \parallel R_1) = (I_b + \Delta I/2)(R_f \parallel R_1)$$

$$V^+ = -I_{b2} \times R_b = -(I_b - I_{os}/2) \times R_b$$

If $R_b = R_f \parallel R_1$:

$$\text{then } \Delta V = V^- - V^+ = R_b \times (I_{b2} - I_{b1}) = R_b \times I_{os}$$

$\therefore$ always present at i/p pins.

$$\text{ie } V_{olp} \text{ error} = (I_{os} \times R_b) \times (1 + R_f/R_1) = I_{os} \times R_f$$

$I_{os} = 5 \text{ pA}$ (datasheet, max 30 pA) $\rightarrow$ if $R_f \uparrow \rightarrow$ o/p error $\uparrow$ (still negligible vs LM741!)

④ O/p Voltage Swing

$\pm 12 \text{ V}$ at $R_L \geq 2 \text{ k}\Omega$, $V_s = \pm 15 \text{ V}$. So o/p is bounded in b/w $\pm 12 \text{ V}$.

⑤ i/p Voltage Range

$\pm 11 \text{ V}$ at 25°C.

⑥ Slew Rate

$SR = 7.5 \text{ V}/\mu\text{s}$ (LT1055 from datasheet).

⑦ GBW

GBW = 4.5 MHz (from datasheet at $f = 1$ MHz test condition).

Layer 2: Resistance Values

$|R_f / R_1| = 100$. So $R_f \approx R_1$ in order of magnitude.

If $R_f \rightarrow M\Omega$, I_b error $\uparrow$ (but still very tiny with JFET input).

If $R_1 = 1\Omega$, $R_f = 100\Omega \rightarrow I_b$ error small, but i/p current too high. LT1055 is JFET so less of an issue, but keep $R_1 = 1k\Omega$, $R_f = 100k\Omega$ for consistency.

Safe (intuitive) range: $R_1 = 1k\Omega$, $R_f = 100k\Omega$

Now due to Input Offset Current $\rightarrow$ Voltage error = $I_{os} \times R_f$:

$$= 5 \text{ pA} \times 100k\Omega = 0.5 \text{ } \mu\text{V} \leftarrow \text{negligible}$$

But due to Input Offset Voltage = $500 \text{ } \mu\text{V}$ (fixed, cannot be removed):

$$\text{Gain error} = 500 \text{ } \mu\text{V} \times 100 = 50 \text{ mV (fixed, cannot be eliminated by resistors).}$$

$$\text{So, } R_b = R_1 || R_f = (1k \times 100k) / (1k + 100k) = 990\Omega \approx 1k\Omega$$

Conclusion (why R_1 , R_f analysis first, not V_{inp} & frequency) :-

Because we always perform static analysis first. Frequency is dynamic, so we set the first analysis when time is not changing — ie the circuit is alive but nothing is moving. In static analysis: R_f , R_b , R_1 | $V_{out}(\text{max})$, $V_{in}(\text{max})$, pk-pk — all are known to us.

Layer 3: Dynamic Analysis

$$f_{3dB} = GBW / |A_v| \quad |A_v| = R_f / R_1 = 100$$

$$GBW = 4.5 \text{ MHz} = 100 \times BW$$

$$\therefore BW = 4.5 \text{ MHz} / 100 = 45 \text{ kHz}$$

$f_{3dB} = 45 \text{ kHz} \rightarrow$ after that frequency, V_{olp} starts attenuating.

Slew Rate Analysis:

$$f_{\text{max}}(\text{SR}) = SR / (2\pi \times V_{\text{out_peak}}) = (7.5 \times 10^6) / (2\pi \times 5) \approx 238 \text{ kHz}$$

$f_{\max}(\text{SR}) \approx 238 \text{ kHz} \rightarrow \text{beyond this frequency, o/p gets distorted.}$

GBW vs SR crossover:

$$f_{\text{GBW}} = f_{\text{SR}} \text{ (crossover condition)}$$

$$\text{GBW} / |A| = \text{SR} / (2\pi \times V_{\text{out_crossover}})$$

$$V_{\text{out_crossover}} = \text{SR} \times |A| / (2\pi \times \text{GBW}) = (7.5 \times 10^6 \times 100) / (2\pi \times 4.5 \times 10^6)$$

$$\mathbf{V_{out_crossover} = 26.5 \text{ V (peak)}}$$

So it is a tradeoff between GBW & slew rate. SR dominates because it changes the signal shape — for amplifier design, signal shape has high priority.

For 5V input signal: $f_{\max}(\text{SR}) \leq 238 \text{ kHz}$, $f_{3\text{dB}} = 45 \text{ kHz} \rightarrow \text{take } 1 \text{ kHz (best here).}$

For 10V input signal: $f_{\max}(\text{SR}) \leq 119 \text{ kHz}$, GBW governs up to 45 kHz $\rightarrow$ again 1 kHz best.

1 kHz can handle both small & large signals.

Layer 4: Power & Thermal Check

$$\text{Power} = I_s \times V_s = 2.8 \text{ mA} \times 30 \text{ V} = 84 \text{ mW}$$

$$\text{Thermal rise} = 84 \text{ mW} \times 150^\circ\text{C/W} = 12.6^\circ\text{C}$$

$$\text{ie } 25^\circ\text{C} + 12.6^\circ\text{C} = 37.6^\circ\text{C} < 150^\circ\text{C} \rightarrow \text{PASS}$$

$$\text{O/p current: } V_{\text{out_max}} / R_L = 5\text{V} / 100 = 0.05 \text{ mA} < 25 \text{ mA} \rightarrow \text{PASS}$$

Final Circuit (at 25°C) :-

$$R_i = 1\text{k}\Omega, R_f = 100\text{k}\Omega, R_b = 1\text{k}\Omega, V_s = \pm 15\text{V}$$

$$V_{\text{inp}} = (0.01 \text{ to } 0.1) \sin(2000\pi t) \text{ V}$$

$$V_{\text{olp}} = -(1 \text{ to } 10) \sin(2000\pi t) \text{ V}$$

LT1055 Inverting Amplifier (Gain = 100) — Simulation

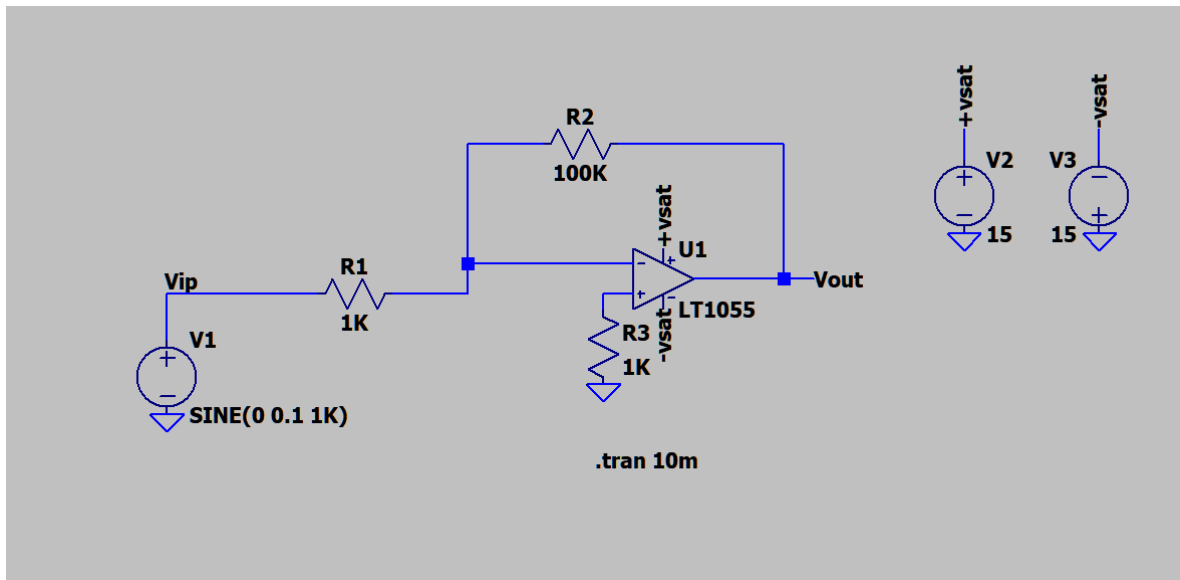


Fig 30: LT1055 LT Spice circuit - same configuration as LM741: $R1 = 1k\Omega$, $R2 = 100k\Omega$, $R3 = 1k\Omega$, $V1 = SINE(0, 0.1, 1000)$, $\pm 15V$ supply, $.tran\ 10m$

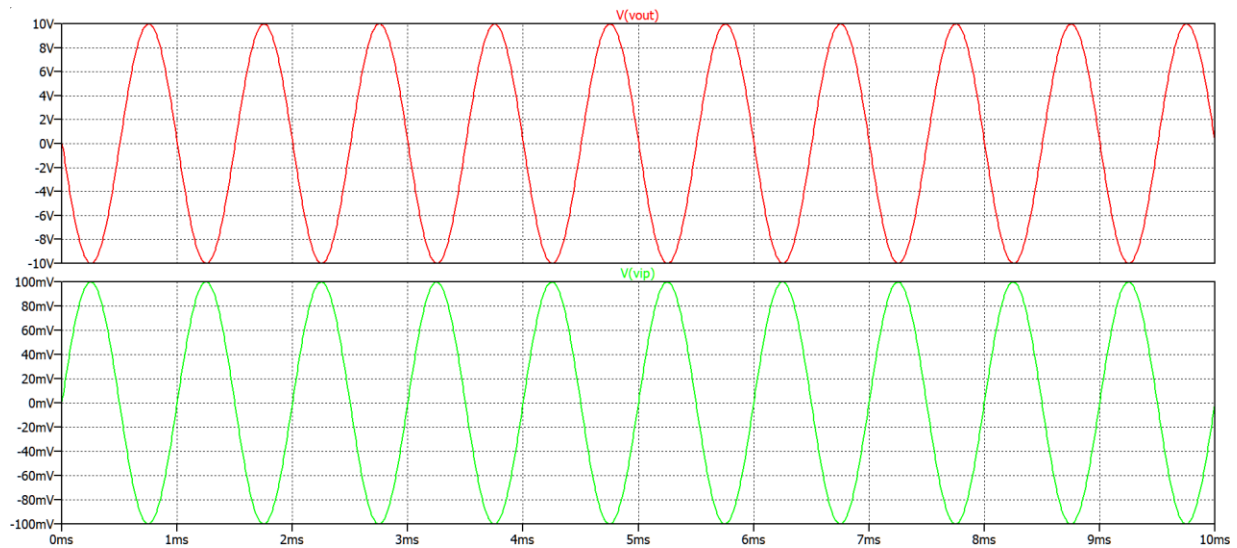


Fig 31: LT1055 transient - $V(vout)$ peak = $\pm 10V$ (red), $V(vip)$ peak = $\pm 100mV$ (green). Gain = 100. Waveform is cleaner than LM741 due to lower distortion

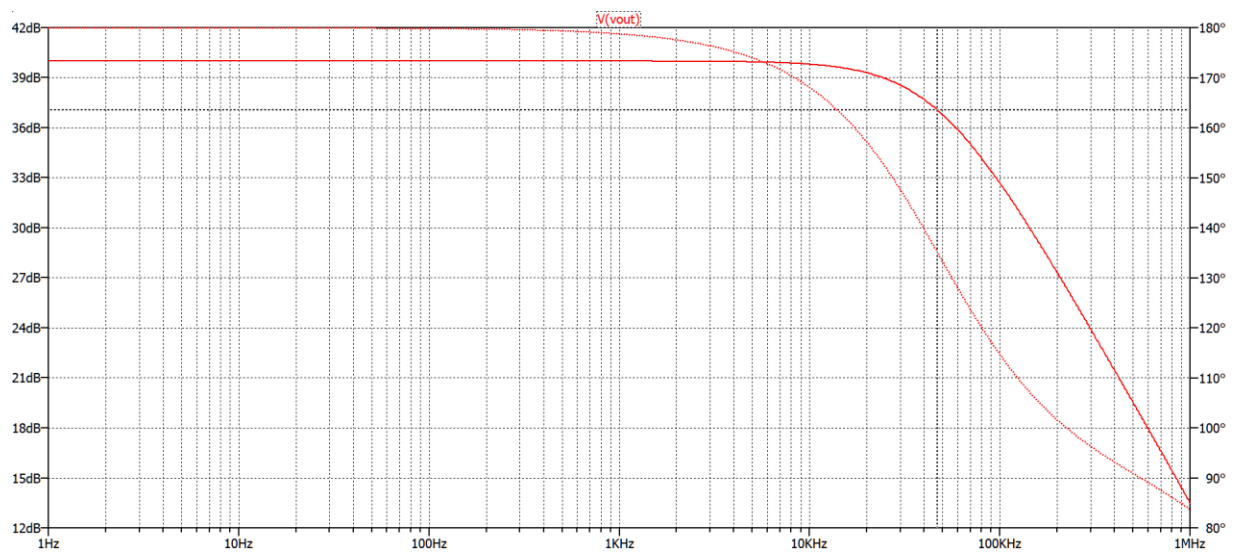


Fig 32: LT1055 AC analysis (Bode plot) - flat gain $\approx 40\text{dB}$ up to $\sim 40\text{kHz}$, then rolls off. Higher bandwidth than LM741 due to $\text{GBW} = 4.5\text{MHz}$

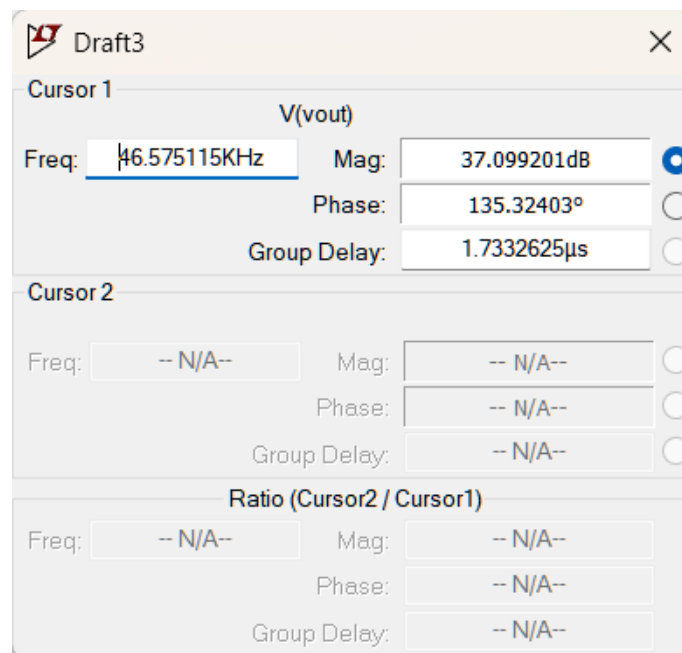


Fig 33: LT1055 Bode cursor at -3dB: $\text{Freq} = 46.58\text{kHz}$, $\text{Magnitude} = 37.1\text{dB} \approx 40 - 3\text{dB}$. $f_{3\text{dB}} \approx 45\text{kHz}$, confirming $\text{GBW}/|A_v| = 4.5\text{MHz}/100 = 45\text{kHz}$

3. Comparison: LM741 vs LT1055

Parameter	LM741 (BJT Input)	LT1055 (JFET Input)
Input Stage Technology	BJT Differential Pair	JFET Differential Pair
Input Bias Current (I_b)	80 nA (typical)	± 30 pA ($\approx 3000\times$ smaller)
Input Offset Voltage (V_{os})	6 mV (max)	500 μ V max ($12\times$ better)
DC Error from Offset Current	$I_{os} \times R_f = 200\text{nA} \times 100\text{k}\Omega = 20$ mV	Negligible (pA-level bias)
DC Error from Offset Voltage	$V_{os} \times (1 + R_f/R_1) = 6\text{mV} \times 101 \approx 606$ mV	~ 50 mV (far lower)
Gain-Bandwidth Product (GBW)	1 MHz	4.5 MHz
-3dB Frequency at Gain = 100	~ 10 kHz	~ 45 kHz
Slew Rate	0.5 V/ μ s	7.5 V/ μ s ($15\times$ faster)
Precision (DC Accuracy)	Low — large offsets cause significant error	High — minimal DC errors
High-Frequency Performance	Limited — slow slew rate, low GBW	Excellent — wide bandwidth, fast slew
Overall Suitability	General purpose, low-cost applications	Precision & high-frequency inverting amp
Cost / Power	Lower cost, lower power	Slightly higher cost and power

Conclusion:

The key distinction between the two op-amps lies in their input stage technology. The LM741 uses a BJT differential pair at the input, resulting in a relatively large input bias current ($I_b = 80\text{nA}$ typ) and significant input offset voltage ($V_{os} = 6\text{mV}$ max). This causes a DC error at the output equal to $I_{os} \times R_f = 200\text{nA} \times 100\text{k}\Omega = 20\text{mV}$ from offset current, plus $V_{os} \times (1 + R_f/R_1) = 6\text{mV} \times 101 \approx 606\text{mV}$ from offset voltage. In contrast, the LT1055 uses a JFET input differential pair, giving an extremely small input bias current ($I_b = \pm 30\text{pA}$, approximately $3000\times$ smaller than LM741) and a much lower input offset voltage ($V_{os} = 500\mu\text{V}$ max, $12\times$ better). As a result, the LT1055 produces far less DC error at the output. On the AC side, the LT1055 has a GBW of 4.5MHz (vs 1MHz for LM741) giving a -3dB frequency of $\sim 45\text{kHz}$ at gain = 100 vs only $\sim 10\text{kHz}$ for LM741, and a slew rate of 7.5 V/ μ s (vs 0.5 V/ μ s) allowing undistorted output up to much higher frequencies and amplitudes. The LT1055 is therefore the clearly superior choice for precision and high-frequency inverting amplifier applications, at the cost of slightly higher price and power.